53 From Point Clouds to Knowledge Graphs: Connecting Archaeological Evidence and Interpretation through FLOSS

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Digital archaeology has often prioritised spatial data acquisition over post-survey interpretation (Landeschi et al. 2018). What outputs should we seek? How far can point clouds be processed and converted into meshes and DEMs without compromising accuracy? How can the chaîne opératoire of knowledge production be made transparent and traceable?

To address these questions, we are testing acquisition techniques, devices and methodological approaches at the late prehistoric site of Ca' Nova di Albareto (Parma Apennines, Italy). Each newly exposed palimpsest surface of interest in Trench A is surveyed using ground- and UAV-based SfM photogrammetry and GSLAM. Individual features are recorded using structured-light scanning and ground-based SfM photogrammetry.

Within the FLOSS environment MAPOD4D® Lab, we produce quality matrices for the point clouds to assess the advantages and limitations of each device. Modules from its MOSAIC subproject, currently under development, support automated noise reduction, calculation of the Global Integrity Index (GII) and construction of reliable models for analysis.

With INFINITY's Knowledge Engineering team, we are exploring a Multi-dimensional Knowledge Graph (mKG) to document excavation data and knowledge production processes. Building on an existing FAIR-compliant database, this approach semantically links stratigraphic records, excavation diaries, artefact documentation, survey data and digital outputs, while recording the methods, observations and interpretative choices underlying them. It aims to make relationships between evidence and interpretation transparent, traceable and reproducible, integrating documentation practices within a shared semantic framework. The intended outcome is an initial prototype for AI-assisted fieldwork documentation.